

Parkbnb V1

Low-Cost Mechanical Architecture & Manufacturing Strategy

Objective

Develop a **low-cost, low-power, robust smart parking lock** using a **mechanically assisted barrier** instead of a continuously motor-driven mechanism.

The design focuses on:

- Minimum BOM Cost (Target: ₹2,000–3,000 Mechanical)
- Easy Manufacturing in Indore
- Low Maintenance
- High Reliability
- Modular Construction
- Standard Off-the-Shelf Components

Design Philosophy

Unlike conventional parking locks where the motor performs the complete barrier movement, **Parkbnb V1 uses mechanical energy for movement while the actuator performs only locking and unlocking.**

Design Rule

Motor/Solenoid → Lock & Unlock Only

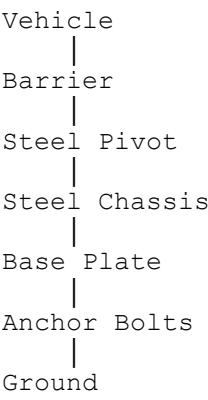
Springs → Motion Assist Only

Steel Structure → Carries All Vehicle Loads

Mechanical Architecture

```
Rubber Protected U Barrier
      |
    Steel Internal Spine
      |
    Dual Pivot Shaft
      |
    Dual Torsion Springs
      |
    Steel Rotary / Hook Lock
      |
    Mini Solenoid
      |
    Steel Chassis
      |
    Steel Base Plate
      |
    Concrete Foundation
```

Vehicle Load Path



Important

Vehicle impact shall **never** be transferred to:

- Springs
 - Solenoid
 - Electronics
 - Battery
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Standard Components (Buy)

Component	Remarks
Torsion Springs	Standard Industrial Spring
Trailer/Rotary Hook Lock	Heavy Duty Steel
Push Solenoid	12V
Hall Sensors	A3144
Magnets	Neodymium
Nylon Bushes	Standard Size
Fasteners	Standard Hardware

Custom Manufactured Parts

Only these parts need fabrication.

Component

Steel Chassis

Base Plate

Side Plates

Pivot Shaft

U Barrier

Lock Bracket

Manufacturing Process

Laser Cutting

- Base Plate
- Side Plates
- Lock Brackets

CNC / Lathe

- Pivot Shaft
- Bush Housing
- Spacer

Welding

- Steel Chassis
- Barrier Frame

Powder Coating

Outdoor corrosion protection

Final Assembly

- Springs
 - Lock
 - Solenoid
 - Sensors
 - Electronics
-

Estimated Mechanical BOM

Component	Approx. Cost
Barrier	₹400
Shaft	₹200
Springs	₹300
Bushes	₹100
Steel Lock	₹350
Solenoid	₹300
Steel Chassis	₹600
Base Plate	₹350
Fasteners	₹100
Hall Sensor	₹50

Mechanical Cost

₹2,300–2,800

Existing Products Used as Inspiration

Instead of developing every component from scratch, the design leverages proven industrial products.

Product	Used For
Motorcycle Center Stand	Pivot & Spring Mechanism
Trailer Over-Center Latch	Locking System
Industrial Torsion Spring	Motion Assist
Nylon Bush	Pivot Support
Steel Floor Enclosure	Base Structure
Gate Spring Hinge	Return Mechanism Reference

Advantages

- Low BOM Cost
 - Low Power Consumption
 - No Heavy Gear Motor
 - Easy Local Manufacturing
 - Modular Design
 - Easy Maintenance
 - High Structural Strength
 - Replaceable Components
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Future Improvements

- Over-Center Linkage Optimization
- Rotary Latch Integration
- Compact Return Mechanism
- Camera-Based Parking Verification
- Ultrasonic Parking Position Detection
- Solar Charging Module
- Predictive Maintenance Dashboard
- OTA Firmware Updates

V1 Target Specifications

Parameter	Target
Mechanical BOM	₹2,000–3,000
Barrier Cycle Time	<5 sec
Power Consumption	Very Low (Lock/Unlock Only)
Operating Life	>100,000 Cycles
Maintenance	Modular & Field Replaceable
Manufacturing	Indore/Pithampur Fabrication

Core USP

A mechanically assisted smart parking lock where electrical power is used only for locking and unlocking, while barrier movement is achieved through optimized mechanical design.

This architecture reduces cost, extends battery life, simplifies maintenance, and is well suited for scalable manufacturing using locally available fabrication capabilities.